



Secure MPC Voting System

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Problem & Objective

Problem Statement

- ▶ Traditional voting systems heavily rely on central trusted authorities.
- ▶ Electronic voting creates single points of failure for data leaks.
- ▶ Voter privacy is at risk during the counting process.
- ▶ Lack of transparency in how votes are aggregated.

Project Objectives

- ▶ Enable counting of votes without ever revealing individual choices.
- ▶ Eliminate the need for a single trusted counting authority.
- ▶ Provide a tamper-resistant digital voting mechanism.
- ▶ Ensure secure and verifiable result aggregation.

Architecture Overview



Local Encryption

Voter encrypts their vote locally on their device. The vote is split into secret shares before leaving the client.



Secure Transmission

Encrypted vote shares are transmitted securely to separate computation servers. No single server holds the full vote.



Joint Computation

Servers jointly compute totals using MPC protocols. The final result is decrypted publicly without exposing inputs.

Gap Analysis

Feature	Traditional / E-Voting	Proposed MPC System
Trust Model	Central Trusted Authority	Distributed Cryptographic Trust
Privacy Risk	High (Authority sees all)	Near Zero
Attack Surface	Central Server (Single Point)	Decentralized
Transparency	Process is Opaque	Verifiable Aggregation

Implementation Roadmap

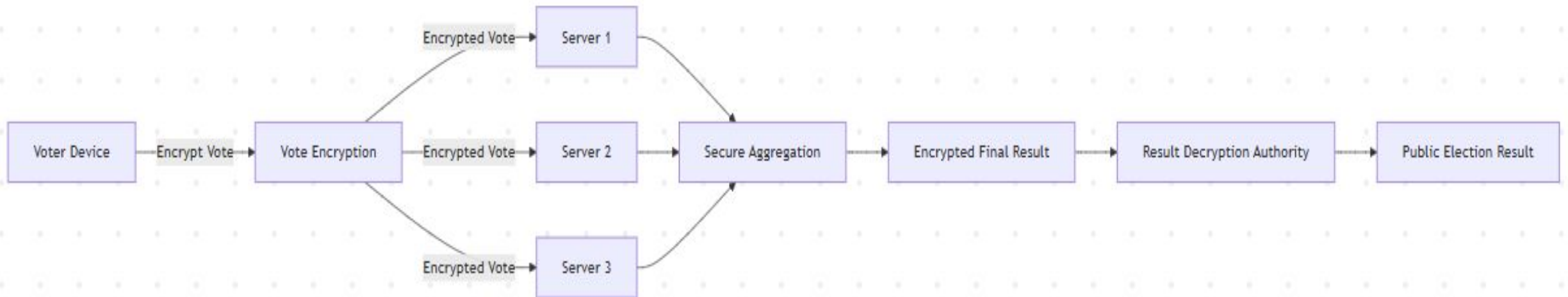


System Architecture

Input: Voters generate secret shares.

Process: Multiple servers compute on shares without reconstruction.

Output: Only the final tally is revealed.



Thank
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